Mihir Agarwal

(646) 235 7363 | mihiragarwal.com | ma4874@columbia.edu | LinkedIn | Github

Education

Columbia University, GPA: 3.67/4 — New York, NY
MS in Data Science (Expected 12/2026) — *Aug 2025 – Dec 2026*
Coursework: Applied Deep Learning, Unsupervised Learning, Statistical Inference, Continual Learning

Vellore Institute of Technology, GPA: 3.82/4 — Vellore, India
Bachelor of Technology (Computer Science) — *July 2018 – July 2022*

Publications

Evaluation benchmark for autonomous database agents : **NeurIPS 2026 (Under Review)**
Memory layers as a RAG replacement in LLMs
ML-based resource management in fog computing : **Springer (IF: 5.2, 37 citations)** DOI

Research Experience

Memory Layers as a RAG Replacement in LLMs — New York, NY
Advisor: Prof. Micah Goldblum | Columbia University — *Jan 2026 – Present*

- Modified Qwen's transformer attention to store document knowledge directly as learned key-value banks inside the model, removing the need for external retrieval (RAG).
- Added MoE-style routing, input-dependent gating, and product-key lookup to control when and how the model reads from its memory layers.
- Pretraining the model end-to-end on 20M+ synthetic QA pairs across science, code, and math, with a multi-stage curriculum that scales from simple retrieval to multi-hop reasoning.
- Deployed inference using vLLM for fast autoregressive decoding during evaluation and benchmarking runs

NeuroLOB: Generative Market Intelligence – Live Demo — New York, NY
Advisor: Prof. Nakul Verma | Columbia University — *Aug 2025 – Dec 2025*

- Designed a hybrid generative framework combining Diffusion models and Causal Transformers with Neural Point Processes (NPP), trained on 19M+ LOB events to simulate asynchronous order book dynamics.
- Outperformed cGAN and NPP baselines in KS fidelity (0.16 vs. 0.28/0.44); higher perplexity (1.68 vs. 1.12) confirmed realistic generation across all LOB event types.

Autonomous Database Agent Evaluation — New York, NY
NeurIPS 2026 (Under Review) | Columbia University — *Sep 2025 – Jan 2026*

- Created an evaluation harness for LLM-based database agents across 4 dimensions most benchmarks ignore: concurrency control, access control, integrity rules, and query efficiency.
- Wrote 433 tasks across 28 risk scenarios from 371 real practitioner incident reports; ran 3-stage quality control with automated checks and manual cross-validation by 4 PhD annotators.
- Benchmarked 8 frontier LLMs in an agentic interaction loop (ReAct) with tool-use orchestration, budget constraints, and LLM-as-judge verification.

Industry Experience

Shell — Bengaluru, India
Machine Learning Engineer — *Aug 2022 – Aug 2025*

- Engineered a production RAG pipeline (LangChain, GPT, FAISS) serving 90,000+ employees at 92% retrieval accuracy, deflecting 68% of HR queries from human agents.
- Deployed as a containerized microservice handling 1,000+ daily requests with P95 latency <1s and $102K in annualized savings.
- Built a test prioritization system using clustering and supervised models over execution history and failure patterns, reducing test cycles by 35%.

Technical Skills

Languages: Python, Java, C/C++, SQL (Postgres), JavaScript, R
ML / AI: PyTorch, JAX, Transformers, BERT, Pretraining, Continual Learning, Curriculum Learning, Sparse MoE, Diffusion Models, Deep RL
LLM Systems: LangChain, LangGraph, RAG, FAISS, ReAct Agents, Tool-Use Orchestration, LLM-as-Judge, vLLM
Infrastructure: Docker, GCP (TPU VMs), Streamlit, FastAPI, Flask, Git, CI/CD